INCH-POUND
MIL-DTL-24509B(SH)
3 October 2017
SUPERSEDING
MIL-V-24509A(SH)
24 December 1992

DETAIL SPECIFICATION

VALVES, FLANGED, BALL AND PLUG FOR SEWAGE AND SEAWATER SERVICE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flanged ball and plug valves for sewage and seawater service, under pressure or vacuum. Sewage contains seawater with varying amounts of detergents, caustics, acids, human wastes, sediments, oils, greases, solvents, and other chemical agents used on board ship.

1.2 Classification. Valves are furnished in the following types, as specified (see 6.2). For flow plans of type II and III valves, see figures 1 through 9.

a. Type I – 2-way, 2-port.

b. Type II – 3-way, 2-port.

c. Type III – 3-way, 3-port.

1.3 Part or identifying number (PIN). The PIN to be used for parts acquired to this specification is constructed as follows:

M	**24509**	**-**	**XX**	**X**	**X**	**X**	**X**
Prefix for Military Specification	Specification Number	-	Size (see code below)	Type (see code below)	Flow Plan (see code below)	Operator (see code below)	Category (see code below)
Example: M24509-25I0LP is for a 2½-inch, type I, two-way, lever operated, plug valve.							

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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Nominal Size (Inches)		Type		Flow Plan		Operator		Category	
Size	Code	Type	Code	Flow	Code	Operator	Code	Category	Code
1¼	12	Type I	I	Two–way	0	Lever	L	Plug	P
1½	15	Type II	II	Figure 1	1	Handwheel	H	Ball	B
2	20	Type III	III	Figure 2	2	Electric actuator	E		
2½	25			Figure 3	3	Other	O		
3	30			Figure 4	4				
3½	35			Figure 5	5				
4	40			Figure 6	6				
4½	45			Figure 7	7				
5	50			Figure 8	8				
6	60			Figure 9	9				
8	80								

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Casting for Seawater Service

DOD-V-24657 - Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)

MIL-C-24723 - Castings, Nickel-Copper Alloy

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-276 - Impregnation of Porous Metal Castings and Powdered Metal Components

MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MILITARY SPECIFICATION SHEETS

MS18229 - Plug for O-Ring Gasket

(Copies of this document are available online at http://quicksearch.dla.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository [NSEDR] online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-RK-STM-010/505 - Naval Ships' Technical Manual, Chapter 505, Piping Systems

(Copies of these documents are available online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering [hard copy] via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads, (UN and UNR Thread Form)

ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

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ASME B18.2.2 - Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A494/A494M - Standard Specification for Castings, Nickel and Nickel Alloy

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B111/B111M - Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock

ASTM B138/B138M - Standard Specification for Manganese Bronze Rod, Bar, and Shapes

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B169/B169M - Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar

ASTM B315 - Standard Specification for Seamless Copper Alloy Pipe and Tube

ASTM B466/B466M - Standard Specification for Seamless Copper-Nickel Pipe and Tube

ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

ASTM D4894 - Standard Specification for Polytetrafluoroethylene (PTFE) Granular Molding and Ram Extrusion Materials

ASTM D7091 - Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals

ASTM F467 - Standard Specification for Nonferrous Nuts for General Use

ASTM F468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS7259 - Rubber: Fluorocarbon (FKM), High Temperature/Fluid Resistant, Low Compression Set/ 85 to 95 Hardness, for Seals in Fuel Systems and Specific Engine Oil Systems

SAE AMS7276 - Rubber: Fluorocarbon (FKM) High-Temperature-Fluid Resistant Low Compression Set for Seals in Fuel Systems and Specific Engine Oil Systems

SAE AMS-STD-595 - Colors Used in Government Procurement

SAE J1926/1 - Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.5.

3.2 Materials. The material for ball and plug valves shall be as specified herein. Material selected shall prevent galling, seizing, and excessive wear of operating parts and shall have galvanic compatibility with system piping. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Stem. Ball valve stems shall be nickel-copper alloy in accordance with QQ-N-281.

3.2.2 Impregnation. Impregnation of valve body castings shall be accomplished in accordance with MIL-STD-276 and S9074-AR-GIB-010/278.

3.2.3 Metallic parts. The bodies, balls, plugs, covers, glands, and base plates shall be made from tin-bronze, alloys C90300 or C92200 in accordance with ASTM B61 or ASTM B584; nickel-aluminum-bronze, alloy C95800 in accordance with ASTM B148 or MIL-B-24480; nickel-copper alloys in accordance with ASTM A494/A494M (grade M-30C, UNS N24130), QQ-N-281, or MIL-C-24723 (composition M-30C); or nickel-copper-aluminum alloy in accordance with QQ-N-286.

3.2.4 Fasteners. Fastener material shall be nickel-copper, grades 400 or 405 in accordance with MIL-DTL-1222, ASTM F467, or ASTM F468 or nickel-copper-aluminum, grade 500 in accordance with MIL-DTL-1222.

3.2.5 O-rings. O-rings shall be fluorocarbon elastomer in accordance with SAE AMS7276 or SAE AMS7259 as appropriate.

3.2.6 Non-metallic parts. Unless otherwise specified herein, seals, seats, gaskets, diaphragms, bearings, and other non-metallic parts shall be commercial, made of fluorocarbon resins or other materials compatible with sewage, seawater, and valve body material. Materials such as glass or graphite filler may be added to fluorocarbon resin to improve wear resistance.

3.2.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Ball and plug valves shall conform to the requirements and dimensions specified herein. Valves shall not leak under positive pressure, atmospheric pressure, or vacuum (where specified) during service use. The valves shall be tight seating at all ports. Valve bodies shall withstand the applied stresses developed during operation of the valve. Valves shall be operable by one person. Plug valves shall not utilize a lift arrangement to unseat the plug before rotating to reduce operating torque. Valves shall not be stick or gun lubricated to effect a seal or reduce operating torque. Low friction fluorocarbon resin material shall be employed on valve surfaces to prevent sticking and to provide low operating torque for ease of actuation. The ball or plug ports shall properly align with the body ports in all flow positions. The plug shall not raise or lower when subject to line pressure or vacuum.

3.3.1 Dimensions. Valve face-to-face and center-to-face dimensions shall be in accordance with table I for the valve size specified (see 6.2). Face-to-face tolerance shall be ±1/16 inch. The center-to-face tolerance shall be ±1/32 inch. Flow passages shall be full ported round bore and shall be equal to the nominal size of the valve (see table I) with a tolerance of +1/16/-0 inch. The inlet and outlet ways shall be chamfered to provide a smoother transition from the valve bore to match the bore of the end flange of the same nominal size. Dimensions and machining of end flanges shall be in accordance with MIL-PRF-20042, class 150. Drilled holes shall straddle the vertical centerline. Spot facing of bearing surfaces for bolt heads and nuts shall be in accordance with MIL-PRF-20042. Valve flange bolt holes may be tapped to suit bolting in areas where valve body interferes with through bolting.

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TABLE I. Dimensions.

Valve Size (Inches)	Face-to-Face 2-Way (Inches)	Center-to-Face 3-Way (Inches)
1¼	5¼	---
1½	5¾	3⅛
2	6½	3½
2½	7¾	4
3	9	4½
3½	10	5
4	10½	5¼
4½	12	---
5	13	6½
6	15	7½
8	19	9½

3.3.2 Rotation. Ball and plug rotation shall be limited by positive stops to the zero and 90-degree positions, and the zero and 180-degree positions, for 90-degree and 180-degree turn valves, respectively (see figures 1 through 9). The position of the ports in the ball or plug shall be shown by permanent and distinct external marks on top of the valve stem. The valve shall incorporate provisions for positively restraining the ball or plug in each of its flow and closed positions.

3.3.3 Stem, ball, and plug. The stem and plug shall be fabricated from one piece, or the stem positively attached to the plug by welding in accordance with S9074-AR-GIB-010/278. Plug and seat surfaces throughout the periphery shall be smooth and provide maximum area contact with minimum clearances. Plug and seat surfaces shall have no grooves, cavities, or relief spaces to trap sediment or solidifying residues between the plug and body during plug rotation and in the plug closed and flow positions. Ball and plug ports shall align with the body in the flow position so as not to create pockets or steps that could collect sewage or solid matter in the valve. Ball valve cavities shall be filled with fluorocarbon resin inserts to prevent trapping sediment or solidifying residues in the valve cavities. Ball valve stems shall not be removable while the valve is installed in the piping and in service, and shall be “blowout” proof in construction, such as with a collared stem. Diverter ball valve stems shall also be designed so the stem can only be installed where the port markings are in alignment with the ball ports.

3.3.4 Low friction surfaces. Valves shall consist of a ball or plug rotating in the valve body cavity and operating against a port seat or seal face of polytetrafluoroethylene (PTFE) material. Valve port seat or seal shall exert positive contact pressure against the ball or plug during rotation and in all valve positions. The valve plug or ball shall be uncoated or coated with PTFE fluorocarbon resin. Low friction treated surfaces shall be either coated or sleeved with fluorocarbon resin (PTFE) in accordance with ASTM D4894. Fluorocarbon resin surfaces that rub on one another shall be of slightly dissimilar formulations to reduce galling. Sleeved plug valves shall consist of a plug rotating in a PTFE resin sleeve positively secured in the valve body plug cavity. The sleeve may also act as the port seal.

3.3.4.1 Coating. Coating material shall be baked on spray-applied PTFE fluorocarbon resin material. Coating thickness shall be a minimum of 1 millimeter (see 4.6.2).

3.3.4.2 Sleeve. Valve plug cavity sleeve shall be molded from PTFE material. The sleeve shall be securely attached in the valve body plug cavity by a suitable locking method such as interlock grips or grooves to prevent loosening in service.

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3.3.5 Handwheels and lever handles. Handwheels smaller than 11 inches in diameter and all levers shall be of commercial design and manufactured from steel, ductile iron, or malleable iron or cast manganese bronzes C86200, C86300, or C86500 in accordance with ASTM B584, or annealed wrought manganese bronzes C67000 or C67500 in accordance with ASTM B138/B138M, or hot formed or annealed aluminum bronzes C61300 or C61400 in accordance with ASTM B169/B169M, or annealed copper-alloy tubing C60800, C61400, C71000, or C71500 in accordance with ASTM B111/B111M, or copper-alloy pipe or tubing C61300 or C61400 in accordance with ASTM B315, or annealed copper-alloy pipe or tubing C71000 or C71500 in accordance with ASTM B466/B466M. Handwheels 11 inches in diameter and larger shall be cast steel in accordance with the general dimensions of 803-1385620. The stem handwheel connection shall be in accordance with 803-1385620 for all sizes. When specified (see 6.2), handwheels and lever handles shall be color coded by painting in accordance with SAE AMS-STD-595.

3.3.6 Handwheel and lever size. Unless otherwise specified (see 6.2), valves under 6 inches shall be furnished with lever handles; valves 6 inches and larger shall be provided with handwheels and shall be operated through enclosed worm and gear mechanisms. Handwheels used to drive manual or electric valve actuators shall be sized to require no more than 80 pounds of tangential force and be no less than 6 inches in diameter. Valve nominal lever lengths and nominal handwheel diameters shall be as specified in table II.

TABLE II. Valve operating torque.

Valve Size (Inches)	Maximum Operating Torque 1/, 2/ (ft/lb)	Nominal Level Length (Inches)	Nominal Handwheel Diameter (Inches)
1¼	20	5	7
1½	25	6	8
2	35	7	11
2½	40	8	11
3	45	9	12
3½	50	10	14
4	60	11	16
4½	70	13	16
5	80	14	18
6	100	--	21
8	120	--	24

NOTES:

1/ Measured at valve stem.

2/ For type II and type III (three-way valves) with more than 2 port seals, maximum operating torque shall be 140% of that shown in table II.

3.3.7 Manual valve actuators. Worm and gear mechanisms shall be enclosed and shall be commercial items except that they shall be shock and vibration tested while mounted on the valve in an operating condition (see 4.6.3 and 4.6.4).

3.3.8 Remote actuating gear. Valves shall be for use with mechanical remote actuating gear in accordance with S9086-RK-STM-010/505. The adapter for remote actuating gear shall be as specified (see 6.2). Valves shall accept remote actuating gear sized to deliver between 20 and 50 percent more torque than the maximum operating torque listed in table II.

3.3.9 Electric valve actuators. When specified (see 6.2), electrically powered valve actuators shall be in accordance with DOD-V-24657. The actuators shall be shock and vibration tested while mounted on the valve in an operating condition (see 4.6.3 and 4.6.4).

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3.3.10 Pipe plugs. Pipe plugs and matching threads, where used, shall be in accordance with MS18229 and SAE J1926/1.

3.3.11 Threaded fasteners. Threads shall be in accordance with ASME B1.1. Bolts and nuts shall be in accordance with ASME B18.2.1 and ASME B18.2.2.

3.4 Performance.

3.4.1 Pressure and temperature rating. Ball and plug valve maximum pressure and temperature rating shall be 150 pounds per square inch (lb/in^2) at 150 °F.

3.4.2 Vacuum rating. Valves 1¼ to 3 inches in size, in addition to operating under pressure (see 3.4.1), shall also operate in vacuum sewage systems at a maximum of 20 inches mercury (Hg).

3.4.3 Shock. Valves shall withstand the shock requirements of grade A, hull mounted, class I, type C in accordance with MIL-DTL-901, and MIL-STD-798 (see 4.6.3).

3.4.4 Vibration. Valves shall withstand the type I vibration requirements in accordance with MIL-STD-167-1 (see 4.6.4).

3.4.5 Structural soundness. Valves shall withstand an internal hydrostatic test pressure in the open position of 225 lb/in^2 for 5 minutes with no leakage, weeping, or structural distortion (see 4.6.5).

3.4.6 Seat tightness (pressure). Valves shall withstand a hydrostatic pressure of 165 lb/in^2 for a minimum of 3 minutes all flow positions with no leakage through the closed port or to the atmosphere (see 4.6.6).

3.4.7 Seat tightness (vacuum). Valves 1¼ to 3 inches in size shall maintain a vacuum of 20 inches of mercury in the closed position for a minimum of 15 minutes with no more than 1 inch of mercury vacuum loss (see 4.6.7). Valves shall meet the above requirement in all flow positions and open position.

3.4.8 Cycle life. Valves shall operate for a life of 2,000 cycles (see 4.6.8).

3.4.9 Operating torque. The maximum allowable valve operating torque, including initial breakaway torque, applied to the valve stem or manual actuator stem, at rated differential pressure (see 3.4.1), or rated vacuum (see 3.4.2), shall be as specified in table II (see 4.6.9 and 4.6.9.1).

3.5 Marking.

3.5.1 Body marking. Valve bodies shall have the pressure rating and manufacturer's name or trademark etched, stamped, or cast integral with the body.

3.5.2 Identification plates. Valves shall bear an identification plate made of corrosion-resisting steel or brass. Identification plates shall be permanently fastened to a part of the valve not subjected to the working pressure. The identification plate shall include the following information:

a. Manufacturer's name or trademark.
b. Size of valve and pressure rating.
c. Body and ball or plug material composition.
d. Manual valve actuator mechanical advantage (if applicable).
e. Manufacturer's identification number or drawing number.
f. Specification MIL-DTL-24509.
g. Allowance parts list (APL) or national stock number (NSN) (if known).
h. Component identification number (CID) (if known).
i. PIN (see 1.3).

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3.6 Special tools. Special tools shall not be required for installation and routine maintenance. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Management Area Operations [DCMAO]). Repair of coated surfaces and sleeve replacement is not considered routine maintenance.

3.7 Workmanship. Valves shall be thoroughly cleaned both inside and out. Valves shall be free from burrs, loose sand, grit, scale, metal shavings, and all other contaminants which can affect performance and damage sealing surfaces. There shall be no sharp edges. All sealing, rubbing, and coated surfaces shall be smooth with no blemishes, scratches, scrapes, or gouges. Machined surfaces shall not exceed 125 roughness height rating (RHR) as defined in ASME B46.1, unless a rougher finish is allowed by specifications. Valve bodies shall meet the visual acceptance criteria for castings in accordance with S9074-AR-GIB-010/278.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein and in 4.6.

4.2.1 Test temperature. Inspections and tests shall be accomplished at ambient room temperature.

4.2.2 Test fluid. All tests shall be conducted using clean fresh water.

4.3 Failures. One or more failures of any test shall be cause for rejection.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in table III.

TABLE III. First article inspection.

Inspection	Requirement	Test
Visual and dimensional examination	3.3.1 through 3.3.11, 3.5.1, 3.5.2, and 3.7	4.6.1
Coating thickness test	3.3.4.1	4.6.2
Shock test	3.4.3	4.6.3
Vibration test	3.4.4	4.6.4
Structural soundness test	3.4.5	4.6.5
Seat tightness (pressure) test	3.4.6	4.6.6
Seat tightness (vacuum) test	3.4.7	4.6.7
Cycle life test	3.4.8	4.6.8
Operating torque test	3.4.9	4.6.9 and 4.6.9.1

4.4.1 First article inspection samples. Unless otherwise specified (see 6.3), one sample of each valve size shall be subjected to the first article inspection specified in 4.4. The first article may be a standard production item from the contractor's current inventory, provided the valve meets the requirements of this specification and is representative of the design, construction, materials, and manufacturing technique applicable to the remaining valves to be furnished under the contract.

4.5 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in table IV.

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4.5.1 Conformance inspection samples. As a minimum, a sample quantity shall be randomly selected from each lot of completed valves as specified in 4.5.3 and inspected as specified in table IV. If one or more defects are found in any sample, the entire lot shall be rejected. The lot may be completely screened for the defective characteristic(s) or a new lot that shall be inspected as specified in 4.6 may be provided.

TABLE IV. Conformance inspection.

Inspection	Requirement	Test
Visual and dimensional examination	3.3.1 through 3.3.11, 3.5.1, 3.5.2, and 3.7	4.6.1
Coating thickness test	3.3.4.1	4.6.2
Structural soundness test	3.4.5	4.6.5
Seat tightness (pressure) test	3.4.6	4.6.6
Seat tightness (vacuum) test	3.4.7	4.6.7
Operating torque test	3.4.9	4.6.9 and 4.6.9.1

4.5.2 Lot. For the purposes of conformance inspection sampling, a lot is defined as all the valves of the same pressure rating, size, and construction, using the same materials and production process, being offered for delivery at one time.

4.5.3 Sampling. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and acceptance quality limit (AQL), as specified (6.2).

4.6 Methods of inspection.

4.6.1 Visual and dimensional examination. Each sample selected as specified in 4.4.1 and 4.5.3 shall be visually and dimensionally examined to determine conformance to this specification, workmanship requirements, and applicable drawings. Major defects shall be classified as shown in table V. Any sample which fails to conform to this visual and dimensional examination shall be cause for rejection of the entire lot.

TABLE V. Classification of defects.

101	Size not as specified.
102	Type not as specified.
103	Material not as specified.
104	Dimensions not as specified.
105	Color not as specified.
106	Component missing or damaged.
107	Improper assembly.
108	Workmanship not as specified.
109	Poor finish on machining.
110	Fluorocarbon resin coating or sleeve rough, blistered, uneven, loose, bubbled, scratched, or gouged.
111	Nominal lever length and nominal handwheel diameter not as specified.
112	Identification markings missing, incorrect, or illegible.

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4.6.2 Coating thickness test. Ball or plug coating thickness shall be measured to determine conformance to 3.3.4.1 using ASTM D7091.

4.6.3 Shock test. Extension of the shock test from one size to other sizes shall be accomplished in accordance with MIL-STD-798. Each valve design shall be shock tested to determine conformance to 3.4.3. Valve actuators, when required, shall be mounted on valves during shock tests (see 3.3.7 and 3.3.9). Where valve actuators are used and extension is to be considered, the largest valve actuator shall be demonstrated on the smallest valve to be included in the shock extension. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. After shock testing, valves shall be tested in accordance with 4.6.6, 4.6.7 as applicable, 4.6.9, and 4.6.9.1.

4.6.4 Vibration test. Each valve design shall be tested to determine conformance to 3.4.4. Valve actuators, when required, shall be mounted on valves during vibration tests (see 3.3.7and 3.3.9). The closed valve shall not open nor shall the open valve close due to vibration. After vibration testing, valves shall be tested in accordance with 4.6.6, 4.6.7 as applicable, 4.6.9, and 4.6.9.1.

4.6.5 Structural soundness test. Each valve shall be hydrostatically tested to determine conformance to the requirements specified in 3.4.5.

4.6.6 Seat tightness (pressure) test. Each valve shall be hydrostatically seat tested in both directions to determine conformance to the requirements specified in 3.4.6.

4.6.7 Seat tightness (vacuum) test. Each valve 1¼ to 3 inches in size shall be vacuum seat tested in both directions and open position to determine conformance to the requirements specified in 3.4.7. Valves shall be tested for vacuum loss using a vacuum volume equivalent to a maximum of 30 feet of schedule 40 pipe of nominal valve size.

4.6.8 Cycle life test. Valves shall be subjected to a cycle test to determine conformance to 3.4.8. A cycle shall consist of the valve ball or plug rotating from one stop position to the other stop position and return. Time for a complete cycle shall not exceed 45 seconds. The valve shall be mounted with the stem horizontal for one-half of the test cycles. The test fluid shall be clean fresh water at ambient temperature. The test shall be conducted at a flow rate of 10 feet per second to simulate hydrodynamic loading. During the cycle test, the stem seal may be adjusted or replaced if necessary not more than four times. The number of cycles between stem seal replacements shall be noted. Upon completion of the cycle test, the valve shall be pressure and vacuum tested and operating torque measured to the requirements specified in 4.6.6, 4.6.7 as applicable, and 4.6.9, and 4.6.9.1, after which the valve shall be internally inspected and results noted. Failure of any of the above tests or any evidence of galling, seizing, peeling, and excessive wear of the coated or sleeved surfaces or the port seals, or the ball or plug, or of any operating parts shall constitute failure of the cycle life test. A loss of 20 percent coating thickness or 0.5 millimeters, whichever is less, constitutes excessive wear.

4.6.9 Operating torque test. Each valve shall be operated and the operating torque measured to determine conformance to 3.4.9. Torques shall be measured at the valve stem with the valve unpressurized, and pressurized to rated differential pressure and rated vacuum. When measuring torque, a gradually increasing applied torque up to three seconds duration is permitted.

4.6.9.1 Operating torque test following a period of non-operation. A differential pressure equal to the rated pressure shall be maintained across the valve for 40 hours. After this period of time has elapsed, the valve shall be fully cycled (closed to open to closed) every 2 hours and the breakaway torque recorded. This shall be continued until such time as three consecutive readings that are both within ±5 percent of each other are obtained. Any torque measurement greater than the maximum operating torque (see table II) shall be cause for rejection.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Valves covered by this specification are high-reliability and are intended for sewage and seawater service aboard Navy ships with a 30-year service life. Valves in sizes specified (see 3.4.2) are suitable for use in vacuum sewage systems. Existing stock of valves manufactured to MIL-V-24509A dated 24 December 1992 are acceptable for use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of valve and flow plan figure number (see 1.2), and whether a ball or plug valve is specifically desired.

c. When first article is required (see 3.1).

d. Valve size (see 3.3.1).

e. When color coding is required and color required (see 3.3.5).

f. Lever or handwheel if other than as specified (see 3.3.6).

g. Remote actuating gear adapter requirements (see 3.3.8).

h. Electric valve actuator, if required (see 3.3.9).

i. Valve sizes for first article inspection where all sizes offered are not selected for test (see 4.4.1).

j. AQL of 1.0 (see 4.5.3).

k. Packaging requirements (see 5.1).

l. Provisioning Technical Documentation (PTD), spare parts, and repair parts, if required (see 6.4).

m. APL or NSN, if available.

6.3 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item should be a first article sample (see 3.1) and the number of items to be tested as specified in 4.5. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to the those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.4 Provisioning. PTD, spare parts, and repair parts should be furnished as specified (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and verifications as the parts used in the manufacture of the equipment. Packing for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contract for incorporation into the equipment and lost their separate identity when the equipment is shipped.

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6.6 Subject term (key word) listing.

CHT

CHT piping system

Coated valve

Collection, holding and transfer

Marine sanitation device

MSD

Plumbing system

Plumbing system valve

Pollution control system

Sewage system valve

Vacuum collection, holding and transfer

VCHT

VCHT piping system

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

NOTE: The following depictions of 3-way, 2-port and 3-way, 3-port ball or plug valves are plan views with the valve stems facing upward. Close attention should be provided as to where the valve stem will be operated from once the valve is installed to prevent operation access issues after installation.

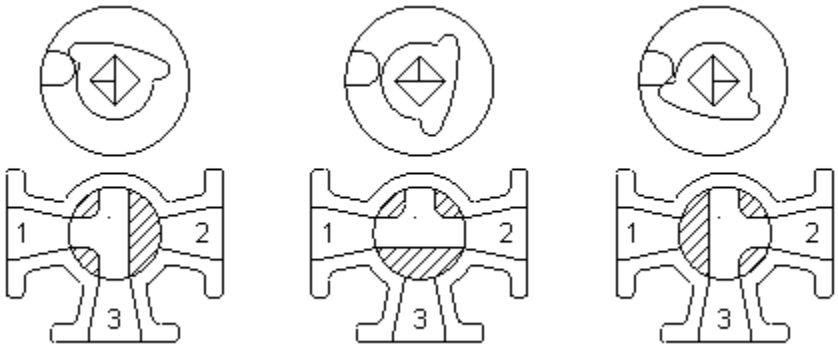


FIGURE 1. 3-way, 3-port, 180° turn.

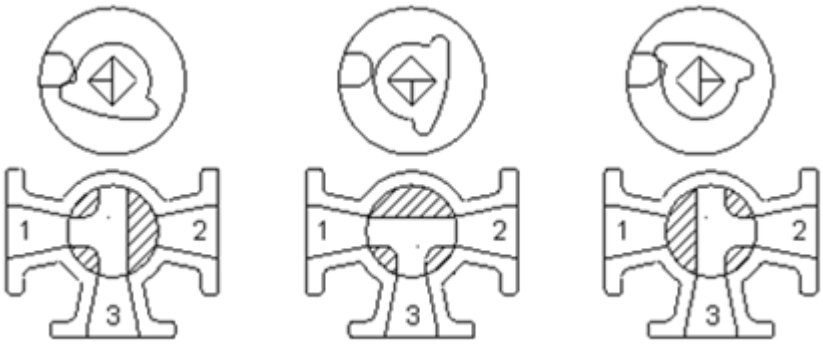


FIGURE 2. 3-way, 3-port, 180° turn.

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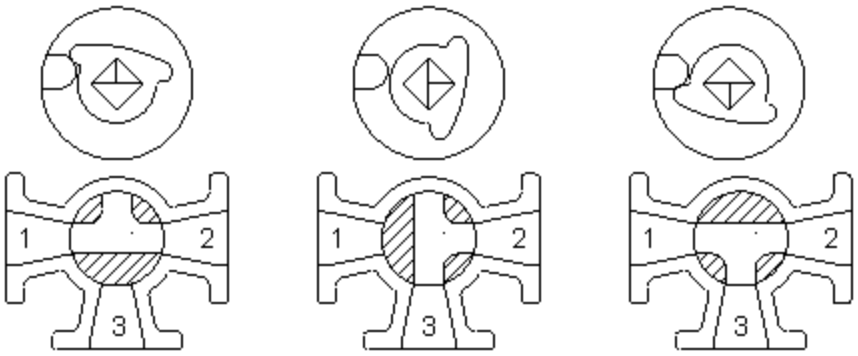


FIGURE 3. 3-way, 3-port, 180° turn.

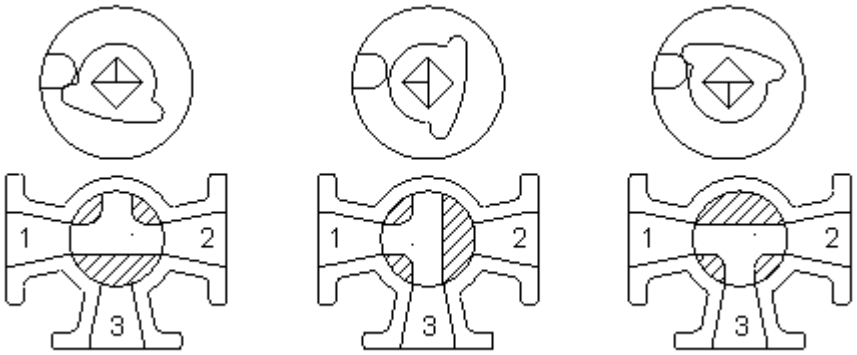


FIGURE 4. 3-way, 3-port, 180° turn.

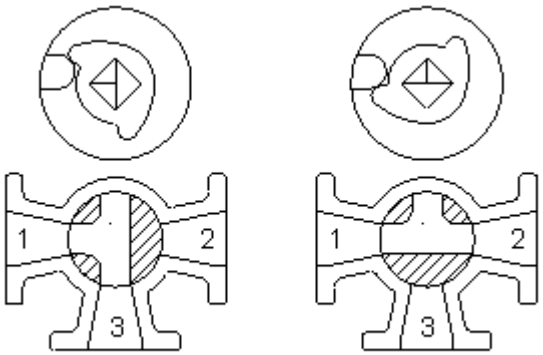


FIGURE 5. 3-way, 3-port, 90° turn.

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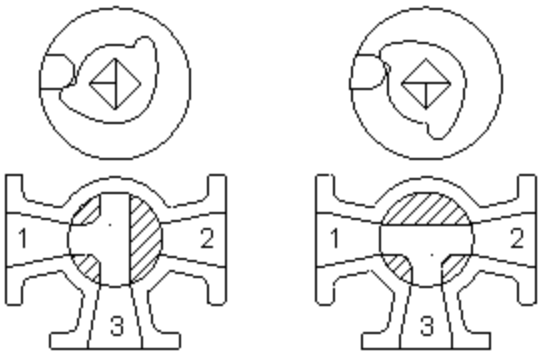


FIGURE 6. 3-way, 3-port, 90° turn.

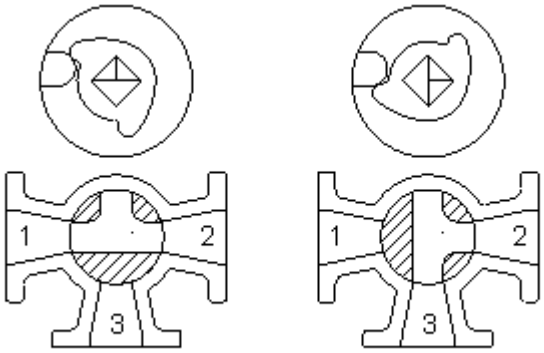


FIGURE 7. 3-way, 3-port, 90° turn.

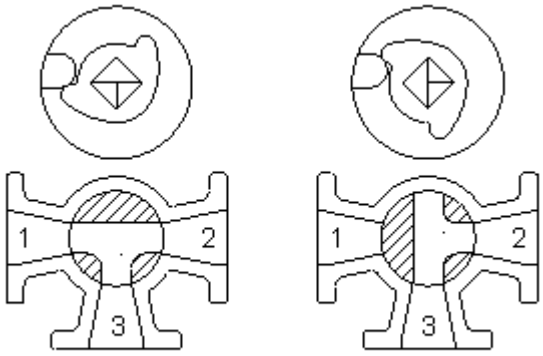


FIGURE 8. 3-way, 3-port, 90° turn.

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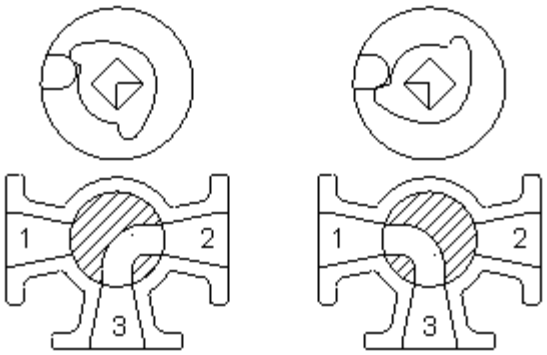


FIGURE 9. 3-way, 2-port, 90° turn.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2017-009)

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